

Exponential Families and the Generalized Linear Model

Every chapter so far has modelled a response whose scatter about a linear mean is, if not normal, at least additive and of constant shape. Counts, proportions, waiting times and insurance claims are none of these: their means are confined to an interval, their spread grows with their mean, their distributions are skewed by construction. This chapter builds the framework that handles them — a response from an **exponential dispersion family**, a **linear predictor** $\mathbf{x}_{(i)}^\top \boldsymbol{\beta}$, and a **link** joining the two. Least squares gives way to maximum likelihood, the normal equations to the likelihood equations, and the closed-form estimate to an iteration that turns out to be weighted least squares, run until it stops changing. Exact distribution theory gives way to asymptotics, so every claim here comes with conditions attached.

What you need. [Chapter 5](#) and [Chapter 6](#) (the linear model, projection, leverage), [Section 7.3](#) and [Section 7.4](#) (maximum likelihood Theorem 7.3.1, the score and information Proposition 7.4.7, exponential families Lemma 7.4.2), [Chapter 11](#) (the likelihood ratio test Theorem 11.4.1 and the trinity Proposition 11.4.2), [Section 21.3](#) (weighted least squares Theorem 21.3.1). [Chapter 20](#) (residuals and influence, and iteratively reweighted least squares Proposition 20.6.1) and [Chapter 29](#) (AIC) are used in the last section, which also needs the chi-squared distribution of a quadratic form from [Chapter 4](#) (Theorem 4.3.2) for the deviance in the normal case.

34.-1.1 Roadmap

- [Exponential dispersion families](#)
- [Link functions](#)
- [The likelihood equations](#)
- [Iteratively reweighted least squares](#)
- [Deviance and asymptotic inference](#)
- [Summary and notes](#)